FoodCal AI

Nutrition Reference Research Report
34 food classes - values per 100 g of prepared edible food

Compiled 11 October 2026

This is a research and data-preparation report. It is not a clinical, laboratory or dietetic analysis, and the values in it are class-level reference compositions rather than measurements of any individual meal.

1. Purpose and scope

FoodCal AI classifies a food photograph into one of 34 predefined classes and then displays energy, protein, carbohydrate and total fat for the identified food. This report establishes the reference nutrition values the application needs, documents where each value came from, and states plainly where a reliable value could not be established.

Of the 34 classes, 29 carry a sourced numeric value and 5 are reported as UNAVAILABLE. No value was invented to complete the set.

2. The per-100 g basis

Every value is expressed per 100 g of edible food as prepared and ready to eat. The application scales to the serving size the user enters:

Serving nutrient = nutrient per 100 g x serving size in grams / 100

For example, a reference of 250 kcal per 100 g gives 250 x 150 / 100 = 375 kcal for a 150 g serving. That is an illustrative arithmetic example; 250 kcal is not a nutrition value for any food in this dataset.

Serving weight must be entered by the user. Nothing in this dataset supports estimating the weight of food from a photograph, and the reference values do not describe the exact composition of any individual photographed meal.

3. How the data was obtained, and an important access limitation

All 29 sourced values come from USDA FoodData Central records - FNDDS survey recipes, SR Legacy, and in three cases Branded Foods. However, **fdc.nal.usda.gov itself could not be reached from the environment used to compile this report**: the site is a JavaScript application that returns no content to a fetch, its public API demo key returned HTTP 429 (rate limited), and the sandbox network blocked the USDA host outright. The UK CoFID publication host was blocked for the same reason.

Each FDC record ID and its nutrient values were therefore read from a public FoodData Central mirror (getfoodfacts.com). This is recorded per record in the verification_status field, which states for every row that the record was read via the mirror and not inspected on the USDA site. The mirror returns genuine FDC identifiers across the SR Legacy, FNDDS and Branded ranges, and one record (chicken curry, FDC ID 2706437) was deliberately re-checked on a second page of the site and agreed exactly. Even so, **before release every value should be re-confirmed directly against fdc.nal.usda.gov**, which takes only a few minutes per record given the IDs listed in section 5.

The Indian Food Composition Tables 2017 were also obtained and inspected to try to fill the remaining Indian-dish gaps. Their 543 records cover raw and basic foods only and contain no prepared dishes, so they could not be used.

4. Nutrition values for all 34 classes (per 100 g)

'n/a' means no reliable reference was established. It does not mean zero.

No	food_class	food_name	preparation_method	Energy kcal	Protein g	Carbs g	Fat g
1	**Baked Potato**	Baked potato (plain, flesh and skin)	Baked, no added fat, no toppings	93	1.95	21.4	0.1
2	**Crispy Chicken**	Crispy fried chicken, breaded boneless pieces	Breaded and deep-fried, boneless, plain (no sauce)	307	15.9	14.9	20.4
3	**Donut**	Doughnut (type not further specified)	Fried sweet dough, type unspecified	426	5.52	49.5	22.9
4	**Fries**	French fries (type not further specified)	Deep-fried potato strips	225	2.5	23.2	14.1
5	**Hot Dog**	Hot dog sandwich on a white bun	Cooked frankfurter in a white bun, no condiments	296	10.8	23.7	17.4
6	**Sandwich**	Sandwich (filling and bread not further specified)	Generic prepared sandwich	287	12.8	33.8	12
7	**Taco**	Beef taco, hard shell	Fried hard corn shell, ground beef, cheese and lettuce	226	8.86	19.9	12.7
8	**Taquito**	Chicken taquito	Rolled filled corn tortilla, chicken filling	269	12.5	31.6	10.4
9	**apple_pie**	Apple pie	Baked double-crust fruit pie, prepared	296	2.7	37.5	15.3
10	**burger**	Hamburger on a white bun (single medium patty)	Grilled beef patty in a white bun, plain	288	17.5	20.7	14.4
11	**butter_naan**	Naan (plain; butter-brushed naan not available)	Leavened tandoor-baked flatbread, plain	311	11.1	50.2	7.28
12	**chai**	Hot tea with milk (sweetened)	Brewed tea with milk and sugar, served hot	51	1.58	8.95	1.05
13	**chapati**	Chapati / roti, whole wheat	Whole-wheat flatbread, commercially prepared (frozen), cooked	299	7.85	46.1	9.2
14	**cheesecake**	Cheesecake, plain	Baked cheesecake, commercially prepared, no topping	321	5.5	25.5	22.5
15	**chicken_curry**	Chicken curry	Prepared chicken curry with sauce, no rice	107	6.48	6.54	6.48
16	**chole_bhature**	Chole bhature (complete dish)	Spiced chickpea curry served with deep-fried leavened bread	n/a	n/a	n/a	n/a
17	**dal_makhani**	Dal makhani (branded ready-meal)	Black gram and kidney bean lentil curry with butter and cream	140	6	14	6.7
18	**dhokla**	Dhokla (steamed savoury gram-flour cake)	Steamed fermented gram-flour batter, tempered	n/a	n/a	n/a	n/a
19	**fried_rice**	Fried rice (type not further specified)	Stir-fried cooked rice with oil	174	3.84	32.5	3.19
20	**ice_cream**	Vanilla ice cream	Frozen dairy dessert, regular fat, vanilla	207	3.5	23.6	11
21	**idli**	Idli (steamed rice and lentil cake)	Steamed fermented rice and black-gram batter	128	6.36	25	0.35
22	**jalebi**	Jalebi (fried batter spirals in sugar syrup)	Deep-fried fermented batter soaked in sugar syrup	n/a	n/a	n/a	n/a
23	**kaathi_rolls**	Kathi roll (approximated by a generic filled wrap)	Filled flatbread wrap	232	12.8	23.3	9.4
24	**kadai_paneer**	Kadai paneer (approximated by paneer makhani)	Paneer cooked in a spiced tomato-based gravy	204	4.9	9.2	16.2
25	**kulfi**	Kulfi (dense Indian frozen dairy dessert)	Reduced sweetened milk, frozen	n/a	n/a	n/a	n/a
26	**masala_dosa**	Dosa with filling	Fermented rice and lentil crepe, griddle-cooked, with filling	184	5.46	30.8	4.27

No	food_class	food_name	preparation_method	Energy kcal	Protein g	Carbs g	Fat g
27	**momos**	Steamed dumplings (momo-style)	Steamed filled dough dumplings	113	6.9	11.1	4.6
28	**omelette**	Plain omelette	Whole eggs cooked as an omelette	154	10.6	0.64	11.7
29	**paani_puri**	Pani puri / golgappa (complete serving)	Crisp hollow puri filled with spiced water, potato and chickpea	n/a	n/a	n/a	n/a
30	**pakode**	Pakora (gram-flour vegetable fritter)	Battered vegetables, fried	125	4.92	16.2	4.77
31	**pav_bhaji**	Pav bhaji - bhaji component only	Spiced mashed vegetable curry with butter (bread NOT included)	136	2.4	8	10.4
32	**pizza**	Cheese pizza, regular crust	Baked, regular crust, cheese topping (pizza-chain reference)	266	11.4	33.3	9.69
33	**samosa**	Samosa	Filled pastry, fried	310	5.14	33.2	17.5
34	**sushi**	Sushi (variety not further specified)	Prepared sushi, variety unspecified	94	2.92	18.4	0.67

5. Source, match type and data quality for each class

No	food_class	Reference record used	Match type	Quality	Source dataset	Record ID	Carbohydrate definition	Verification status
1	**Baked Potato**	Potato, baked, NFS	DIRECT_MATCH	HIGH	FNDDS	2709383	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
2	**Crispy Chicken**	Fast foods, chicken, breaded and fried, boneless pieces, plain	DIRECT_MATCH	HIGH	SR Legacy	170718	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
3	**Donut**	Doughnut, NFS	DIRECT_MATCH	MEDIUM	FNDDS	2708062	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
4	**Fries**	Potato, french fries, NFS	DIRECT_MATCH	HIGH	FNDDS	2709456	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
5	**Hot Dog**	Hot dog sandwich, NFS, on white bun	DIRECT_MATCH	HIGH	FNDDS	2707056	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
6	**Sandwich**	Sandwich, NFS	DIRECT_MATCH	MEDIUM	FNDDS	2706880	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
7	**Taco**	Fast foods, taco with beef, cheese and lettuce, hard shell	DIRECT_MATCH	MEDIUM	SR Legacy	170689	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
8	**Taquito**	Taquito, chicken	DIRECT_MATCH	MEDIUM	FNDDS	2708601	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
9	**apple_pie**	Pie, apple	DIRECT_MATCH	HIGH	FNDDS	2707995	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
10	**burger**	Hamburger, on white bun, 1 medium patty	DIRECT_MATCH	HIGH	FNDDS	2706932	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
11	**butter_naan**	Bread, naan	APPROXIMATE_MATCH	MEDIUM	FNDDS	2707613	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
12	**chai**	Tea, hot, with milk	APPROXIMATE_MATCH	MEDIUM	FNDDS	2710506	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
13	**chapati**	Bread, chapati or roti, whole wheat, commercially prepared, frozen	DIRECT_MATCH	MEDIUM	SR Legacy	174075	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
14	**cheesecake**	Cake, cheesecake, commercially prepared	DIRECT_MATCH	HIGH	SR Legacy	172711	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
15	**chicken_curry**	Chicken curry	DIRECT_MATCH	MEDIUM	FNDDS	2706437	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
16	**chole_bhature**	-	UNAVAILABLE	NONE	-	-	-	requires research
17	**dal_makhani**	Dal Makhani - Mother India (branded)	APPROXIMATE_MATCH	LOW	Branded Foods	1862742	Total carbohydrate as declared on the manufacturer's US Nutrition Facts label (includes fibre); label-rounded	source record read via fdc mirror; some values arithmetically scaled from declared serving
18	**dhokla**	-	UNAVAILABLE	NONE	-	-	-	requires research
19	**fried_rice**	Rice, fried, NFS	DIRECT_MATCH	MEDIUM	FNDDS	2708952	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site

No	food_class	Reference record used	Match type	Quality	Source dataset	Record ID	Carbohydrate definition	Verification status
20	**ice_cream**	Ice cream, vanilla	DIRECT_MATCH	HIGH	FNDDS	2705630	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
21	**idli**	Idli	DIRECT_MATCH	HIGH	FNDDS	2708346	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
22	**jalebi**	-	UNAVAILABLE	NONE	-	-	-	requires research
23	**kaathi_rolls**	Sandwich wrap, NFS	APPROXIMATE_MATCH	LOW	FNDDS	2706881	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
24	**kadai_paneer**	Paneer Makhani - Haldiram Snacks Pvt. Ltd. (branded)	APPROXIMATE_MATCH	LOW	Branded Foods	484203	Total carbohydrate as declared on the manufacturer's US Nutrition Facts label (includes fibre); label-rounded	source record read via fdc mirror; some values arithmetically scaled from declared serving
25	**kulfi**	-	UNAVAILABLE	NONE	-	-	-	requires research
26	**masala_dosa**	Dosa, with filling	APPROXIMATE_MATCH	MEDIUM	FNDDS	2709129	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
27	**momos**	Wonton, dumpling or pot sticker, steamed	APPROXIMATE_MATCH	MEDIUM	FNDDS	2708708	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
28	**omelette**	Egg, whole, cooked, omelet	DIRECT_MATCH	HIGH	SR Legacy	172185	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
29	**paani_puri**	-	UNAVAILABLE	NONE	-	-	-	requires research
30	**pakode**	Pakora	DIRECT_MATCH	MEDIUM	FNDDS	2710066	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
31	**pav_bhaji**	Authentic Indian Mashed Vegetable Curry Pav Bhaji - Kitchens of India (branded)	APPROXIMATE_MATCH	LOW	Branded Foods	2677881	Total carbohydrate as declared on the manufacturer's US Nutrition Facts label (includes fibre); label-rounded	source record read via fdc mirror; some values arithmetically scaled from declared serving
32	**pizza**	Fast Food, Pizza Chain, 14 inch pizza, cheese topping, regular crust	DIRECT_MATCH	HIGH	SR Legacy	173292	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
33	**samosa**	Samosa	DIRECT_MATCH	MEDIUM	FNDDS	2708730	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site
34	**sushi**	Sushi, NFS	DIRECT_MATCH	MEDIUM	FNDDS	2708959	Carbohydrate, by difference (USDA convention; includes fibre)	source record read via fdc mirror not inspected on usda site

6. Source record URLs

The canonical URL is the authoritative USDA address for the record and is where the value should be re-confirmed. The mirror URL is the page that was actually read when compiling this report.

food_class	Record ID	Canonical USDA URL (authoritative, not accessed)	Mirror URL (actually read)
Baked Potato	2709383	https://fdc.nal.usda.gov/food-details/2709383/nutrients	https://getfoodfacts.com/food/potato-baked-nfs-2709383
Crispy Chicken	170718	https://fdc.nal.usda.gov/food-details/170718/nutrients	https://getfoodfacts.com/food/fast-foods-chicken-breaded-and-fried-boneless-pieces-plain-170718
Donut	2708062	https://fdc.nal.usda.gov/food-details/2708062/nutrients	https://getfoodfacts.com/food/doughnut-nfs-2708062
Fries	2709456	https://fdc.nal.usda.gov/food-details/2709456/nutrients	https://getfoodfacts.com/food/potato-french-fries-nfs-2709456
Hot Dog	2707056	https://fdc.nal.usda.gov/food-details/2707056/nutrients	https://getfoodfacts.com/food/hot-dog-sandwich-nfs-on-white-bun-2707056
Sandwich	2706880	https://fdc.nal.usda.gov/food-details/2706880/nutrients	https://getfoodfacts.com/food/sandwich-nfs-2706880
Taco	170689	https://fdc.nal.usda.gov/food-details/170689/nutrients	https://getfoodfacts.com/food/fast-foods-taco-with-beef-cheese-and-lettuce-hard-shell-170689
Taquito	2708601	https://fdc.nal.usda.gov/food-details/2708601/nutrients	https://getfoodfacts.com/food/taquito-chicken-2708601
apple_pie	2707995	https://fdc.nal.usda.gov/food-details/2707995/nutrients	https://getfoodfacts.com/food/pie-apple-2707995
burger	2706932	https://fdc.nal.usda.gov/food-details/2706932/nutrients	https://getfoodfacts.com/food/hamburger-on-white-bun-1-medium-patty-2706932
butter_naan	2707613	https://fdc.nal.usda.gov/food-details/2707613/nutrients	https://getfoodfacts.com/food/bread-naan-2707613
chai	2710506	https://fdc.nal.usda.gov/food-details/2710506/nutrients	https://getfoodfacts.com/food/tea-hot-with-milk-2710506
chapati	174075	https://fdc.nal.usda.gov/food-details/174075/nutrients	https://getfoodfacts.com/food/bread-chapati-or-roti-whole-wheat-commercially-prepared-frozen-174075
cheesecake	172711	https://fdc.nal.usda.gov/food-details/172711/nutrients	https://getfoodfacts.com/nutrition/cheesecake
chicken_curry	2706437	https://fdc.nal.usda.gov/food-details/2706437/nutrients	https://getfoodfacts.com/food/chicken-curry-2706437
dal_makhani	1862742	https://fdc.nal.usda.gov/food-details/1862742/nutrients	https://getfoodfacts.com/food/dal-makhani-1862742
fried_rice	2708952	https://fdc.nal.usda.gov/food-details/2708952/nutrients	https://getfoodfacts.com/food/rice-fried-nfs-2708952
ice_cream	2705630	https://fdc.nal.usda.gov/food-details/2705630/nutrients	https://getfoodfacts.com/food/ice-cream-vanilla-2705630
idli	2708346	https://fdc.nal.usda.gov/food-details/2708346/nutrients	https://getfoodfacts.com/food/idli-2708346
kaathi_rolls	2706881	https://fdc.nal.usda.gov/food-details/2706881/nutrients	https://getfoodfacts.com/food/sandwich-wrap-nfs-2706881
kadai_paneer	484203	https://fdc.nal.usda.gov/food-details/484203/nutrients	https://getfoodfacts.com/food/paneer-makhani-484203
masala_dosa	2709129	https://fdc.nal.usda.gov/food-details/2709129/nutrients	https://getfoodfacts.com/food/dosa-with-filling-2709129
momos	2708708	https://fdc.nal.usda.gov/food-details/2708708/nutrients	https://getfoodfacts.com/food/wonton-dumpling-or-pot-sticker-steamed-2708708
omelette	172185	https://fdc.nal.usda.gov/food-details/172185/nutrients	https://getfoodfacts.com/food/egg-whole-cooked-omelet-172185
pakode	2710066	https://fdc.nal.usda.gov/food-details/2710066/nutrients	https://getfoodfacts.com/food/pakora-2710066
pav_bhaji	2677881	https://fdc.nal.usda.gov/food-details/2677881/nutrients	https://getfoodfacts.com/food/authentic-indian-mashed-vegetable-curry-pav-bhaji-2677881

food_class	Record ID	Canonical USDA URL (authoritative, not accessed)	Mirror URL (actually read)
pizza	173292	https://fdc.nal.usda.gov/food-details/173292/nutrients	https://getfoodfacts.com/food/fast-food-pizza-chain-14-pizza-cheese-topping-regular-crust-173292
samosa	2708730	https://fdc.nal.usda.gov/food-details/2708730/nutrients	https://getfoodfacts.com/food/samosa-2708730
sushi	2708959	https://fdc.nal.usda.gov/food-details/2708959/nutrients	https://getfoodfacts.com/food/sushi-nfs-2708959

7. Recipe and preparation limitations, class by class

Every class carries limitations. Classes marked APPROXIMATE_MATCH or LOW quality should be shown to users with a visible caveat in the application.

food_class	Match / quality	Limitations	Notes
Baked Potato	DIRECT_MATCH HIGH	Reference is a plain baked potato with no added fat. Butter, sour cream, cheese or bacon toppings commonly shown in food photographs are NOT included and would raise energy and fat substantially.	NFS = 'not further specified' (USDA generic survey code). Near-zero fat confirms no added fat in the reference.
Crispy Chicken	DIRECT_MATCH HIGH	Boneless breaded pieces (tender/nugget style). Bone-in, skin-on fried chicken and sauce-coated or sandwich forms differ. Source does not state skin-on/skinless.	Record is explicitly breaded and fried and plain, matching the 'crispy chicken' class well.
Donut	DIRECT_MATCH MEDIUM	NFS generic: icing, glaze and filling are not specified. Related USDA records range from 363 kcal (custard-filled) to 434 kcal (cake type, plain) per 100 g, so a specific doughnut may differ from this generic value by roughly +/-15%.	Chosen as the generic entry because the classifier returns only 'Donut' without type.
Fries	DIRECT_MATCH HIGH	Generic NFS record. Fat reflects deep-frying oil absorption. Restaurant and fast-food USDA records are higher (289 and 312 kcal/100 g); a school-service record is lower (166 kcal/100 g). Toppings (cheese, chili) are excluded.	Oven-baked home fries would be lower in fat than this reference.
Hot Dog	DIRECT_MATCH HIGH	Includes BOTH sausage and bun. Ketchup, mustard, relish, onions and chili are NOT included; the USDA chili hot dog records are 219-223 kcal/100 g. Turkey and reduced-fat variants are lower (237-248 kcal/100 g).	Mirror page header rounded protein to 10.9 g; the detailed nutrient table value 10.8 g was used.
Sandwich	DIRECT_MATCH MEDIUM	'Sandwich' is an inherently heterogeneous class. This generic NFS record should not be treated as representative of any one recipe: USDA sandwich records range from 232 kcal/100 g (wrap, NFS) to 382 kcal/100 g (peanut butter).	Deliberately a generic record, since no single sandwich recipe is universal.
Taco	DIRECT_MATCH MEDIUM	Shell type and filling are fixed to hard-shell beef. Soft-shell beef versions are 206-219 kcal/100 g; chicken, fish and vegetarian fillings were not evaluated. Sour cream, guacamole and salsa are not included.	The narrow 206-229 kcal/100 g spread across USDA taco records makes this a reasonably stable reference for the class.
Taquito	DIRECT_MATCH MEDIUM	Filling fixed to chicken. USDA sibling records: beef or pork 287, egg 281, cheese only 309 kcal/100 g. The record name does not state whether the taquito was deep-fried or oven-baked, so frying oil uptake is not verifiable from the source.	Chicken chosen as the most common commercial taquito filling.
apple_pie	DIRECT_MATCH HIGH	Prepared pie including pastry, as required. Record does not distinguish home-baked from commercially prepared, and does not include a la mode ice cream or cream.	Explicitly a prepared pie record, not raw apples.
burger	DIRECT_MATCH HIGH	Includes BOTH patty and bun. Cheese, bacon, mayonnaise and other sauces are NOT included - a cheeseburger would be higher in fat and energy. Patty size variants (small/large/double) differ only slightly per 100 g (280-290 kcal/100 g).	Per-100 g energy is stable across USDA hamburger size variants, which supports using this as the class reference.
butter_naan	APPROXIMATE_MATCH MEDIUM	KEY LIMITATION: this is PLAIN naan, not butter naan. No USDA record for butter- or ghee-brushed naan was found. Butter naan carries additional surface butter/ghee, so true values will be HIGHER in fat and energy than shown. The size of that gap is not quantified by any source consulted, so no adjustment has been applied.	Searches for 'butter naan' in FoodData Central returned only unrelated branded items (butter chicken naan bites), none of which is a butter naan bread.
chai	APPROXIMATE_MATCH MEDIUM	Not specifically Indian masala chai. The record reports 8.72 g total sugars out of 8.95 g carbohydrate, which indicates the reference IS sweetened, but the milk-to-water ratio and spice content are unspecified. Indian chai is typically brewed with a higher proportion of whole milk and would generally be higher in fat and energy.	Sugar content was checked specifically to confirm the reference is a sweetened milk tea.
chapati	DIRECT_MATCH MEDIUM	Commercially prepared frozen product rather than a home-made chapati. At 9.2 g fat per 100 g this reference is noticeably richer than a typical home roti made without added oil or ghee, so it may overstate fat and energy for home-cooked chapati. The USDA plain (non-whole-wheat) chapati record is 297 kcal/100 g.	Whole-wheat version chosen because Indian chapati is made from whole-wheat atta.
cheesecake	DIRECT_MATCH HIGH	Plain commercially prepared cheesecake. Fruit toppings, chocolate, caramel and whipped-cream garnishes are not included. Style (New York / baked / no-bake) is not specified by the record.	Record identified via the mirror's generic cheesecake page, which cites FDC ID 172711.

food_class	Match / quality	Limitations	Notes
chicken_curry	DIRECT_MATCH MEDIUM	FNDDS survey recipe; the record does not state whether it represents home-cooked or restaurant/takeaway preparation, nor the regional style. At 107 kcal/100 g it is lower than many restaurant curries, which use more oil and cream. Excludes rice (the separate USDA 'Chicken curry with rice' record is 116 kcal/100 g).	Protein and fat are both 6.48 g; this coincidence was re-checked on a second page of the mirror and is what the source reports.
chole_bhature	UNAVAILABLE NONE	No composition record was found for the COMPLETE dish. Searches for 'chole', 'bhatura' and 'chana masala' in FoodData Central returned no bhatura/bhature record at all, and only chickpea-curry components. Per the project brief, chickpea curry alone must not be presented as the complete dish, and the fried bread contributes a large share of the energy, so no value is reported.	Components located but NOT used: 'Punjabi Choley' (branded, FDC ID 484207, 162 kcal/100 g) for the curry, and 'Bread, puri' (FNDDS, FDC ID 2707714, 409 kcal/100 g) as a fried-bread proxy. A composite could be built from these two if a serving ratio is chosen, but that ratio would be an assumption, not sourced data.
dal_makhani	APPROXIMATE_MATCH LOW	MANUFACTURER-SPECIFIC branded product, not a universal composition for dal makhani. Submitted ingredients do include butter and cream, matching the dish. Carbohydrate and fat were arithmetically scaled from the declared 150 g serving values (21 g and 10 g), so they inherit label rounding. Other branded dal makhani products in FDC range from 105 to 225 kcal/100 g, which shows how much recipe variation exists. Restaurant dal makhani typically uses more butter and cream than a retail ready-meal.	No generic (non-branded) USDA record for dal makhani exists; the generic USDA 'Dal' record (FDC ID 2707427, 145 kcal/100 g) lacks the butter and cream that define this dish.
dhokla	UNAVAILABLE NONE	No composition record for PREPARED steamed dhokla was found. The only FoodData Central match is a dry instant mix ('Deep, Indian Gourmet Dhokla Mix', branded, 360 kcal/100 g), which the project brief explicitly rules out, since a dry mix is not the ready-to-eat steamed food.	Entries exist on community calorie-counter sites, but those lack documented analytical methodology and permissive reuse terms, so they were not used.
fried_rice	DIRECT_MATCH MEDIUM	NFS generic: egg, meat and vegetable content are not specified by the record. USDA sibling records are tightly clustered (meatless 174, chicken 173, pork 179, beef 178, shrimp 166 kcal/100 g), so the generic value is a reasonable class reference. Restaurant fried rice made with more oil would be higher.	The narrow spread across protein variants supports use of the NFS record.
ice_cream	DIRECT_MATCH HIGH	Flavour fixed to vanilla and fat level is regular. Premium/rich ice creams, light and reduced-fat products, and chocolate or mix-in flavours differ. Cones, sauces and toppings are not included.	Vanilla selected as the standard reference flavour; SR Legacy carries an equivalent record (FDC ID 167575) with the same 207 kcal/100 g.
idli	DIRECT_MATCH HIGH	Steamed preparation, as required. Accompaniments (sambar, coconut chutney, podi, ghee) are NOT included and would add substantially to fat and energy in a typical served portion. The near-zero fat (0.35 g/100 g) confirms a plain steamed reference.	Record is the prepared food, not a dry idli batter or mix.
jalebi	UNAVAILABLE NONE	No record found. Searches for 'jalebi' and the alternate spelling 'jilebi' in FoodData Central returned no matches at all, generic or branded. The Indian Food Composition Tables 2017 (IFCT, 543 records) were also checked and contain only raw and basic foods, no prepared sweets.	Remaining gap: a composition record for prepared jalebi that captures both frying oil uptake and sugar-syrup absorption. Community calorie-counter entries exist but have no documented methodology or clear reuse licence.
kaathi_rolls	APPROXIMATE_MATCH LOW	KEY LIMITATION: this is a generic filled wrap, NOT a kathi roll. A kathi roll uses a paratha or roti that is typically pan-fried in oil, often with an egg coating, and is filled with spiced chicken, mutton or paneer. The USDA generic wrap is tortilla-based and its filling is unspecified, so this reference most likely UNDERSTATES fat and energy. Treat the number as indicative only.	No kathi roll, frankie or comparable Indian roll record exists in FoodData Central; a shawarma search returned only one branded lavash wrap, which is no closer a match.
kadai_paneer	APPROXIMATE_MATCH LOW	KEY LIMITATION: this is paneer MAKHANI, a butter- and cream-based paneer curry, not kadai paneer, which is built on onion, tomato and capsicum with less dairy fat. The reference therefore probably OVERSTATES fat. It is also a single manufacturer's recipe. Carbohydrate and fat were scaled from the declared 142 g serving values (13 g and 23 g) and inherit label rounding.	Used in preference to plain paneer cheese (FDC ID 2705740, 299 kcal/100 g), which is an ingredient rather than the prepared dish. No kadai paneer record exists in FoodData Central.
kulfi	UNAVAILABLE NONE	No usable record found. FoodData Central searches returned a dry dessert mix (Gits Kesar Kulfi Dessert Mix, 400 kcal/100 g, not a prepared food) and two kulfi bar products whose individual record pages could not be opened to read macronutrients, so their IDs could not be confirmed. Ordinary ice cream was deliberately NOT substituted: kulfi is made from reduced milk, is denser and is not churned.	Remaining gap: a verifiable prepared-kulfi record with full macronutrients.
masala_dosa	APPROXIMATE_MATCH MEDIUM	The record is a FILLED dosa, which is the right category, but USDA does not state that the filling is potato masala specifically. For contrast the plain dosa record is 210 kcal/100 g (FDC ID 2708347). Sambar and chutney served alongside are not included.	Chosen over plain dosa because the brief requires a filled dosa reference where available.

food_class	Match / quality	Limitations	Notes
momos	APPROXIMATE_MATCH MEDIUM	Generic East Asian steamed dumpling rather than a Nepali/Tibetan momo, and the filling is unspecified. Cooking method IS steamed, as required. FRIED momos would be substantially higher in fat and energy and are not represented by this record. The low energy density reflects the high moisture content of steamed dumplings.	No generic USDA momo record exists; branded momo products carried no confirmable cooking method.
omelette	DIRECT_MATCH HIGH	Plain omelette. Cheese, vegetables, onions, meat and masala are NOT included; USDA treats those as separate ingredient records. The source page gives no explicit note on cooking fat, but the fat level (11.7 g/100 g versus about 10 g for raw whole egg) is consistent with a modest amount of added fat.	An Indian masala omelette with onion, chilli and extra oil would differ from this reference.
paani_puri	UNAVAILABLE NONE	No record found for the complete dish; searches for 'pani puri' in FoodData Central returned no matches. Plain puri was deliberately NOT substituted: the USDA 'Bread, puri' record (FDC ID 2707714) is 409 kcal/100 g, whereas a served pani puri is dominated by spiced water and would have a far lower energy density. Using the bread alone would overstate energy by a large margin.	Remaining gap: a composition record for an assembled pani puri serving including the spiced water and fillings.
pakode	DIRECT_MATCH MEDIUM	FLAGGED FOR REVIEW: at 125 kcal and 4.77 g fat per 100 g this record is markedly lower than values commonly cited for Indian deep-fried pakoda, which are usually in the 250-320 kcal/100 g range. The USDA survey recipe may represent a preparation that absorbs much less oil, or a higher-moisture vegetable fritter. The record's own energy and macronutrients are internally consistent, so this is a recipe difference rather than a data error. Verify against your own use case before relying on it.	The record name matches the dish exactly, but the preparation behind the survey recipe is not documented on the source page.
pav_bhaji	APPROXIMATE_MATCH LOW	KEY LIMITATION: this is the BHAJI ONLY. The pav (buttered bread rolls) is not included, even though the product name contains 'Pav Bhaji'. A complete pav bhaji serving includes bread and additional butter and would have a different composition. It is also a single manufacturer's recipe. Carbohydrate and fat were scaled from the declared 125 g serving values (10 g and 13 g) and inherit label rounding.	Submitted ingredients (potato, tomato, bell pepper, peas, butter, cream) confirm it is a genuine bhaji. If your app shows the complete dish, this value will understate it.
pizza	DIRECT_MATCH HIGH	Cheese topping and regular crust specifically. Meat toppings (pepperoni, sausage), thin or stuffed crusts, and extra-cheese versions differ. The comparable frozen cooked cheese pizza record is 268 kcal/100 g, which is close.	Restaurant/pizza-chain reference chosen as most representative of photographed pizza.
samosa	DIRECT_MATCH MEDIUM	The USDA record does not state the filling, so it cannot be confirmed as the vegetable (potato and pea) samosa that the brief specifies as the intended reference; meat-filled samosas would differ in protein and fat. Chutneys are not included. The fat level (17.5 g/100 g) is consistent with deep-frying.	This is the only generic samosa record in FoodData Central.
sushi	DIRECT_MATCH MEDIUM	NFS generic: the variety is not specified, and sushi varieties differ greatly. The very low fat (0.67 g/100 g) indicates the reference does NOT represent tempura rolls, mayonnaise-dressed rolls or fatty-fish nigiri, all of which would be considerably higher in fat and energy. Soy sauce is not included.	Per the brief, this value should not be assumed to apply to all sushi types.

8. Data quality summary

Metric	Count
Total food classes	**34**
Direct matches	**21**
Approximate matches	**8**
Unavailable entries	**5**
High-quality records	**11**
Medium-quality records	**14**
Low-quality records	**4**
No quality rating (no value reported)	**5**
Records requiring further research	**5**
Classes with a numeric energy value	**29**

Energy and macronutrient consistency check

Each record was checked against the diagnostic estimate 4 x protein + 4 x carbohydrate + 9 x fat. All 29 records with values agree with their reported energy to within 3.2%, which is well inside the range explained by fibre, rounding, specific energy factors and differing carbohydrate definitions. No source energy value was replaced by the formula and no record was adjusted to fit it.

One value flagged for review

pakode. The USDA 'Pakora' record gives 125 kcal and 4.77 g fat per 100 g, well below the 250-320 kcal per 100 g commonly cited for Indian deep-fried pakoda. The record's own energy and macronutrients are internally consistent, so this is a difference in the underlying survey recipe rather than a data error - it may describe a high-moisture fritter that absorbs far less oil. Confirm it suits your use case before relying on it.

9. Unavailable entries and research gaps

food_class	What was searched for, and what is still missing
chole_bhature	No composition record was found for the COMPLETE dish. Searches for 'chole', 'bhatura' and 'chana masala' in FoodData Central returned no bhatura/bhature record at all, and only chickpea-curry components. Per the project brief, chickpea curry alone must not be presented as the complete dish, and the fried bread contributes a large share of the energy, so no value is reported. Components located but NOT used: 'Punjabi Choley' (branded, FDC ID 484207, 162 kcal/100 g) for the curry, and 'Bread, puri' (FNDDS, FDC ID 2707714, 409 kcal/100 g) as a fried-bread proxy. A composite could be built from these two if a serving ratio is chosen, but that ratio would be an assumption, not sourced data.
dhokla	No composition record for PREPARED steamed dhokla was found. The only FoodData Central match is a dry instant mix ('Deep, Indian Gourmet Dhokla Mix', branded, 360 kcal/100 g), which the project brief explicitly rules out, since a dry mix is not the ready-to-eat steamed food. Entries exist on community calorie-counter sites, but those lack documented analytical methodology and permissive reuse terms, so they were not used.
jalebi	No record found. Searches for 'jalebi' and the alternate spelling 'jilebi' in FoodData Central returned no matches at all, generic or branded. The Indian Food Composition Tables 2017 (IFCT, 543 records) were also checked and contain only raw and basic foods, no prepared sweets. Remaining gap: a composition record for prepared jalebi that captures both frying oil uptake and sugar-syrup absorption. Community calorie-counter entries exist but have no documented methodology or clear reuse licence.

food_class	What was searched for, and what is still missing
kulfi	No usable record found. FoodData Central searches returned a dry dessert mix (Gits Kesar Kulfi Dessert Mix, 400 kcal/100 g, not a prepared food) and two kulfi bar products whose individual record pages could not be opened to read macronutrients, so their IDs could not be confirmed. Ordinary ice cream was deliberately NOT substituted: kulfi is made from reduced milk, is denser and is not churned. Remaining gap: a verifiable prepared-kulfi record with full macronutrients.
paani_puri	No record found for the complete dish; searches for 'pani puri' in FoodData Central returned no matches. Plain puri was deliberately NOT substituted: the USDA 'Bread, puri' record (FDC ID 2707714) is 409 kcal/100 g, whereas a served pani puri is dominated by spiced water and would have a far lower energy density. Using the bread alone would overstate energy by a large margin. Remaining gap: a composition record for an assembled pani puri serving including the spiced water and fillings.

These five classes carry null values in the JSON dataset and blank cells in the workbook. They must never be rendered as zero. The application should show that nutrition data is unavailable for the class instead.

10. Licensing and redistribution

USDA FoodData Central. The data are produced by a US Government agency and are in the public domain in the United States; USDA asks that the database be cited. Redistribution inside an application is generally permitted on that basis. Because the USDA terms page could not be opened from the compiling environment, this status is stated from general knowledge and is **marked for verification** before public release.

Branded Foods records. The three branded records (dal_makhani, kadai_paneer, pav_bhaji) are manufacturer-declared label values distributed through FoodData Central. Brand and product names are manufacturer trademarks. Do not present a branded product as the universal composition of a dish; the dataset labels each one as manufacturer-specific and LOW quality.

The mirror site. getfoodfacts.com was used as an access route only. Its own terms of use were not reviewed and are **marked for verification**. Nothing of the site's own content is reproduced in these deliverables - only the underlying USDA values and record identifiers.

UK CoFID. Not used. An Open Government Licence is typical for that publication but was not verified, since the host was unreachable. Note that CoFID reports AVAILABLE carbohydrate, which is not the same definition as USDA carbohydrate by difference; if CoFID records are added later, the difference must be recorded rather than silently mixed.

Indian Food Composition Tables 2017. Obtained as npm package 'ifct2017' v2.1.2 and inspected, but not used, as it contains no prepared dishes. Underlying data rights were not verified.

Community calorie-counter sites. Entries for jalebi, dhokla and kulfi exist on such sites, but they are largely user-contributed, carry no documented analytical methodology and have restrictive terms. They were rejected as sources, which is why those classes are reported as UNAVAILABLE.

11. Final verification checklist

Check	Result
Exactly 34 food classes are represented	Yes - verified programmatically; 34 rows, 34 unique identifiers
All class identifiers match the supplied list exactly	Yes - compared element by element against the brief, including order, spelling and capitalization
Every numeric value has an identifiable source	Yes - each carries a dataset name and an FDC record ID
All values expressed per 100 g	Yes - per 100 g edible portion throughout
Correct units (kcal, g) and total fat rather than saturated fat	Yes - energy in kcal, macronutrients in g, total lipid used
Preparation methods documented	Yes - a preparation_method field for every class
Direct and approximate matches distinguished	Yes - 21 direct, 8 approximate, 5 unavailable
Missing values are null or blank, never zero	Yes - 5 classes carry nulls in JSON and blank cells in the workbook
Source URLs and record IDs are genuine	IDs and values were read from a public FDC mirror and are reported as such; canonical USDA URLs are given for independent checking but were NOT accessed - see section 3
Licensing limitations disclosed	Yes - section 10, with two items explicitly marked for verification
Carbohydrate definitions recorded	Yes - by difference for USDA records, label total carbohydrate for the three branded records
Major energy/macronutrient discrepancies flagged	Yes - all 29 records agree within 3.2%; pakode flagged separately on recipe grounds
The JSON parses successfully	Yes - re-parsed after writing; 34 records, no NaN or Infinity
The workbook opens and recalculates cleanly	Yes - recalculated with LibreOffice; 12 formulas, 0 errors; summary counts verified

Check	Result
All three files are consistent with one another	Yes - all three are generated from one shared dataset definition
No claim of verification made without evidence	Yes - every record states that it was read via a mirror and not inspected on the USDA site

This report documents research and data preparation. It is not a clinical or laboratory analysis, and the match-type and quality ratings are research assessments rather than formal validation.